



01 INTRODUCTION

Priyanka is a Biotechnology undergraduate at Lovely Professional University who works comfortably at the intersection of biology, code, and hardware. Her project record spans an automated street-light system built on an AVR microcontroller and a soil-less aquaponic growing platform, alongside 40+ hours of community development fieldwork reaching over 900 people. She brings a builder's instinct to problems — prototyping with sensors and relays as readily as she plans and executes a community outreach program — and is drawn to work where scientific thinking meets practical, sustainable impact.

02 ACADEMIC BACKGROUND

AUG 2025 — PRESENT

Bachelor of Biotechnology

Lovely Professional University, Punjab, India • CGPA: 7.73

MAR 2023 — MAY 2024

Intermediate (PCM)

LT G.S. Salaria Memorial Govt. Senior Secondary School, Punjab, India • 92.8%

JUL 2021 — MAY 2022

Matriculation

St. Andrews Convent School, Punjab, India • 79%

03 SKILLS

LANGUAGES

C

C++

Python

SOFT SKILLS

Time Management

Problem Solving

Teamwork

Leadership

Creative Thinking

04 INTERNSHIP

Community Development Project

Jun — Jul 2026

Summer Intern — Community Services & Social Welfare

- Completed 40+ hours of hands-on training in community development, applying community services and social welfare practices across multiple real-world tasks.
- Implemented community services and social welfare initiatives using a community development model, supporting 900+ users and improving community welfare.
- Executed end-to-end outreach workflows including field surveys, community interactions, data collection of 900+ records, and impact evaluation to improve program efficiency.

05 PROJECTS

Smart Street Light for Energy Conservation

Apr 2025

AVR • Arduino IDE • C++ • Sensors

- Engineered an automated street light system using an AVR microcontroller to monitor real-time vehicle movement based on sensor data.
- Designed threshold-based control logic with relay switching to light up automatically as vehicles pass, preventing over-lighting.
- Assembled a hardware prototype with a sensor, relay module, and DC supply for automatic lighting in road applications.

Aquaponic System — Soil-less Plant Cultivation

Feb 2025

Sensors • Relay Modules • DC Water Pump

- Conceived an automated aquaponic platform for growing plants without soil with the help of fish, supporting sustainable agriculture.
- Designed pump control logic with relay switching to circulate water automatically between fish tank and plants.
- Assembled a working prototype with a soil moisture sensor, relay module, and DC water pump for automatic irrigation, reducing manual effort.

Cross-Disciplinary Builder

Moves fluidly between biology coursework and hands-on electronics — sensors, relays, and microcontrollers.

Structured Problem Solver

Approaches projects with clear time management, breaking down complex builds into testable steps.

Community-Oriented Leadership

Led outreach touching 900+ people, translating leadership and teamwork into measurable, real-world results.

Creative, Sustainability-Minded

Consistently steers projects — from aquaponics to smart lighting — toward resource-conscious design.

Python Programming Certificate

Infosys

May 2025

Certificate in Critical Thinking and Logic

Saylor Academy

Apr 2026

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To grow as a biotechnologist who builds — combining lab-grounded scientific knowledge with hands-on engineering to design sustainable, community-focused solutions, and to keep learning across disciplines rather than within just one.